

SCH 3111



By

Dr. Bosire

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Periodic table

1 1 H 1.008 ← Atomic number, Z ← Element symbol ← Relative atomic mass, A_r																																				18	
1 H 1.008		2 He 4.00																																			
3 Li 6.94		4 Be 9.01																		13 B 10.81		14 C 12.01		15 N 14.01		16 O 16.00		17 F 19.00		18 Ne 20.18							
11 Na 22.99		12 Mg 24.31																		13 Al 26.98		14 Si 28.09		15 P 30.97		16 S 32.06		17 Cl 35.45		18 Ar 39.95							
19 K 39.10		20 Ca 40.08		21 Sc 44.96		22 Ti 47.90		23 V 50.94		24 Cr 52.01		25 Mn 54.94		26 Fe 55.85		27 Co 58.93		28 Ni 58.69		29 Cu 63.54		30 Zn 65.41		31 Ga 69.72		32 Ge 72.59		33 As 74.92		34 Se 78.96		35 Br 79.91		36 Kr 83.80			
37 Rb 85.47		38 Sr 87.62		39 Y 88.91		40 Zr 91.22		41 Nb 92.91		42 Mo 95.94		43 Tc 98.91		44 Ru 101.07		45 Rh 102.91		46 Pd 106.42		47 Ag 107.87		48 Cd 112.40		49 In 114.82		50 Sn 118.71		51 Sb 121.75		52 Te 127.60		53 I 126.90		54 Xe 131.30			
55 Cs 132.91		56 Ba 137.34		La-Lu		72 Hf 178.49		73 Ta 180.95		74 W 183.85		75 Re 186.21		76 Os 190.23		77 Ir 192.22		78 Pt 195.08		79 Au 196.97		80 Hg 200.59		81 Tl 204.37		82 Pb 207.19		83 Bi 208.98		84 Po 210		85 At 210		86 Rn 222			
87 Fr 223		88 Ra 226.03		Ac-Lr		104 Rf [261]		105 Db [262]		106 Sg [266]		107 Bh [264]		108 Hs [277]		109 Mt [268]		110 Ds [271]		111 Rg [272]		112 Cn [285]		113 Uut [284]		114 Uuq [289]		115 Uup [288]		116 Uuh [291]		118 Uuo [294]					
Lanthanoids				57 La 138.91		58 Ce 140.12		59 Pr 140.91		60 Nd 144.24		61 Pm 146.92		62 Sm 150.35		63 Eu 151.96		64 Gd 157.25		65 Tb 158.92		66 Dy 162.50		67 Ho 164.93		68 Er 167.26		69 Tm 168.93		70 Yb 173.04		71 Lu 174.97					
Actinoids				89 Ac 227.03		90 Th 232.04		91 Pa 231.04		92 U 238.03		93 Np 237.05		94 Pu 239.05		95 Am 241.06		96 Cm 244.07		97 Bk 249.08		98 Cf 252.08		99 Es 252.09		100 Fm 257.10		101 Md 258.10		102 No 259		103 Lr 262					



BASIC INORGANIC AND PHYSICAL CHEMISTRY



Recommended Textbooks

Petrucci R. H. and Harwood W. S., "General Chemistry: Principles and Modern Applications".

McMurry J. and Fay R. C., "Chemistry". Prentice Hall. New Jersey.

Raymond Chang, "Essential Chemistry". McGraw-Hill Companies, Inc. Toronto

Chemistry:- John McMurry

Atkins, *et al.*, (2010) Inorganic Chemistry

Inorganic Chemistry by Catherine E. Housecroft
& Alan G. Sharpe (4th Edition)

Learn

Read

Acquire Knowledge

Never forget the **MAIN** reason you have come to campus !!



More Textbooks

1. Clugston, M. and Flemming, R., *Advanced Chemistry*, Oxford University Press.
2. Clark, J., (2014) *Calculations in AS/A Level Chemistry*, Pearson Education Limited.
3. Ramsden, E.N., (2000) *A-Level Chemistry*, Oxford University Press.

....Etc



Course Content and Outline



- 1) **Atomic Theory:** Atomic structure, electron configuration, orbitals
- 2) **Periodic table & Chemical bonding:** Types of chemical bonds, oxidation-reduction; solubility and precipitation; Chemistry of carbonates (Sodium and calcium); silica; manufacture of acids, ammonia; cement and glass.
- 3) **Ideal gas laws:** Kinetic theory of gases, viscosity, vapor pressure, surface tension.
- 4) **Thermochemistry:** Enthalpy and Entropy, Hess law
- 5) **Chemical equilibrium:** Equilibrium law, Equilibrium constant, homogeneous equilibrium, heterogeneous equilibrium, Lechatelier's principle; ionic equilibrium, solubility product.
- 6) **Chemical kinetics:** Rate of reactions and rate equations, factors that influence rate of reaction, order of reaction, molecularity of reaction, activation energy.
- 7) **Adsorption:** Types of adsorption, adsorption isotherms, applications of adsorption.
- 8) **Electrochemistry:** Corrosion and Pourbaix diagrams

**Genius is
1% talent and
99% hard work.**

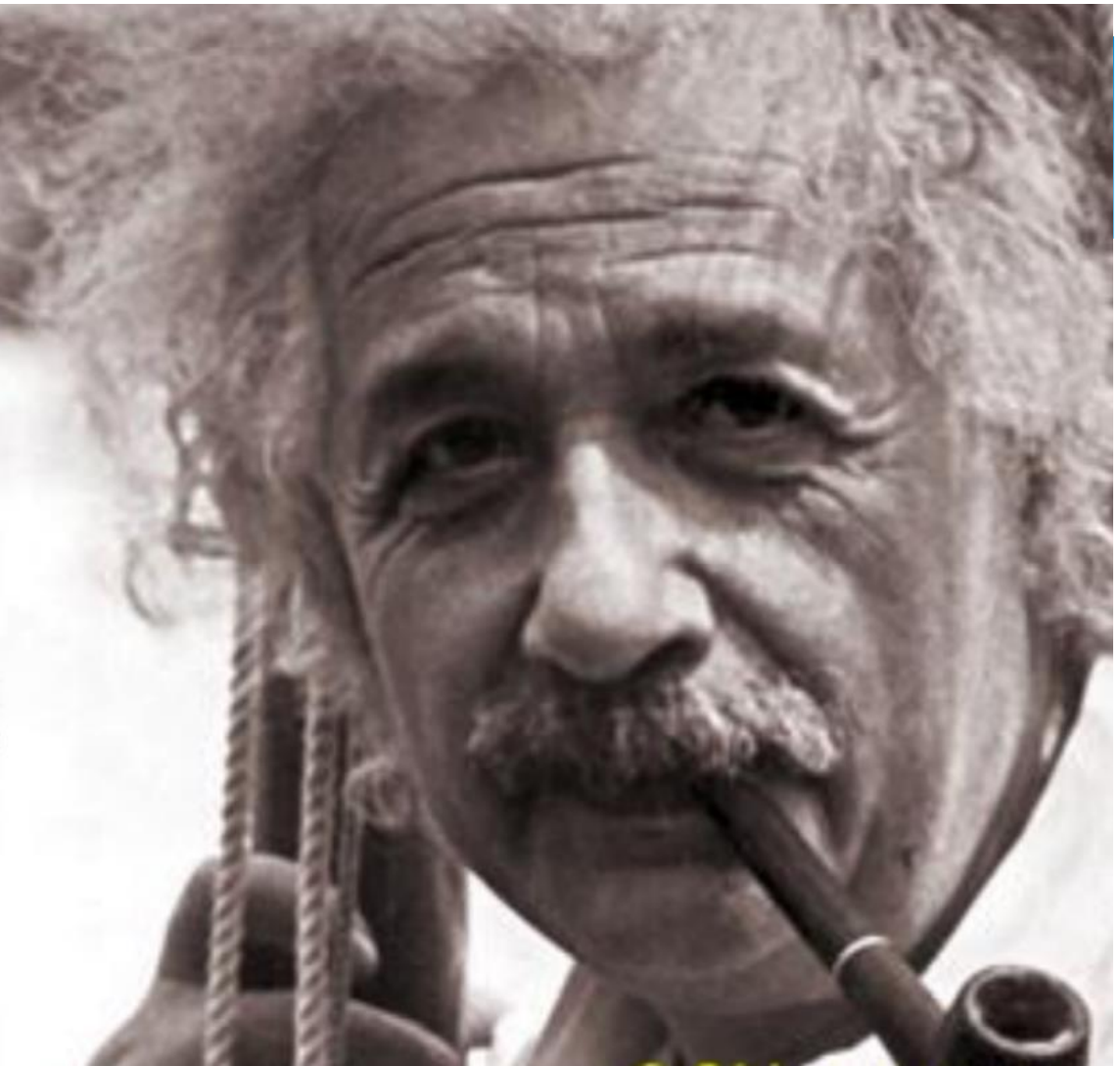
Albert Einstein

1% Driver Genius

**You want to be a great
Engineer?**



*Success Depends on
the Second Letter*





Lecture 1



Structure of the Atom

An atom is the smallest unit quantity of an element that is capable of existence, either alone or in chemical combination with other atoms of the same or another element.

The fundamental particles of which atoms are composed are the **proton, electron and neutron**.



Classical View of Atomic Structure



The fundamental particles are:

Electrons (Symbol e^-)	Protons (Symbol p)	Neutrons (Symbol n)
Discovered by J. J. Thomson in 1897 , when conducting experiment on conductivity of electricity of gases at very low pressure. He initially called them cathode rays .	Protons were discovered by J. J. Thomson , when later conducting experiment on conductivity of electricity of gases at very low pressure. He initially called them positive rays	Discovered by Chadwick in 1932 . From his experiments he showed that neutrons had almost the same mass as the protons. He explained the difficulty concerning the relative atomic masses of hydrogen and helium and later on the idea of isotopes .

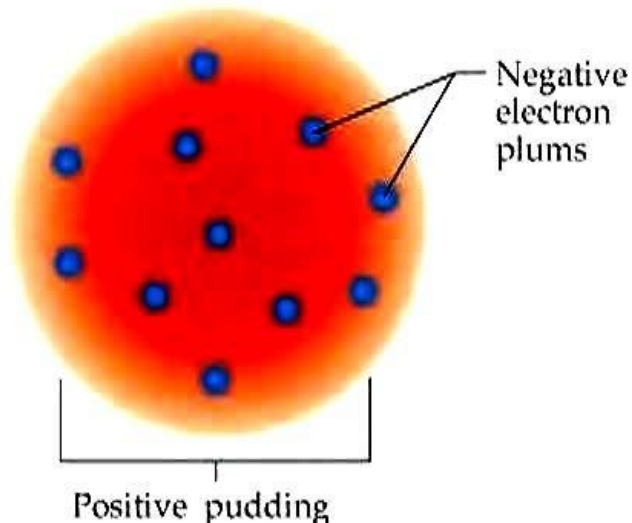
	Proton	Electron	Neutron
Charge /C	$+1.602 \times 10^{-19}$	-1.602×10^{-19}	0
Charge number (relative charge)	1	-1	0
Rest mass/kg	1.673×10^{-27}	9.109×10^{-31}	1.675×10^{-27}
Relative mass	1837	1	1839



Historical Development of Atomic Theory

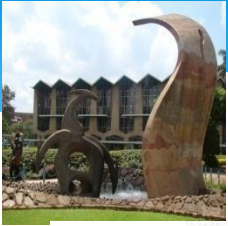


Thomson's Plum Pudding or Raisin Bun Atomic Model (1856-1940)



Thomson suggested a positive charge that is uniformly distributed in an electron embedded sphere in a way that attraction of the *positive charge offsets the repulsion among the electrons*. This somehow resembles raising pudding.

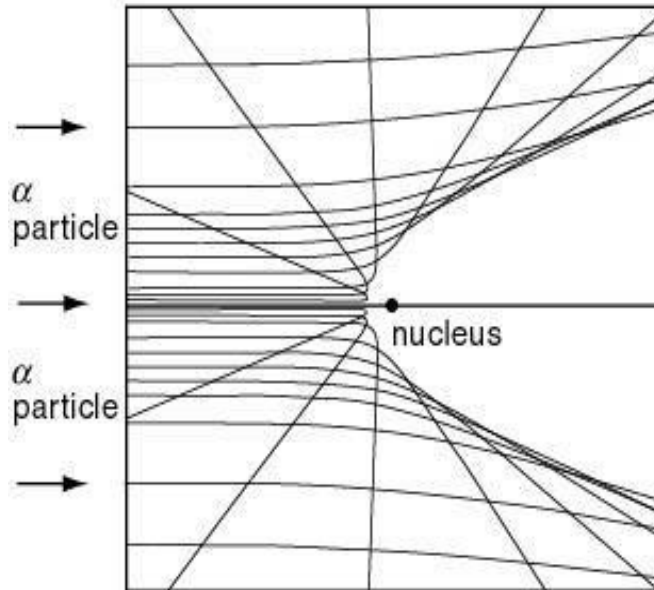
Thomson's plum pudding or raisin bun atomic model



Rutherford's Nuclear Model of Atom (1871-1937)



Trajectories of alpha particles



Angular Distribution of Rutherford Scattering

Bombarded a very thin foil of metal such as **gold, silver and platinum** with α -particles.

They found that;

i) Most of the particles went right through the foil undeflected

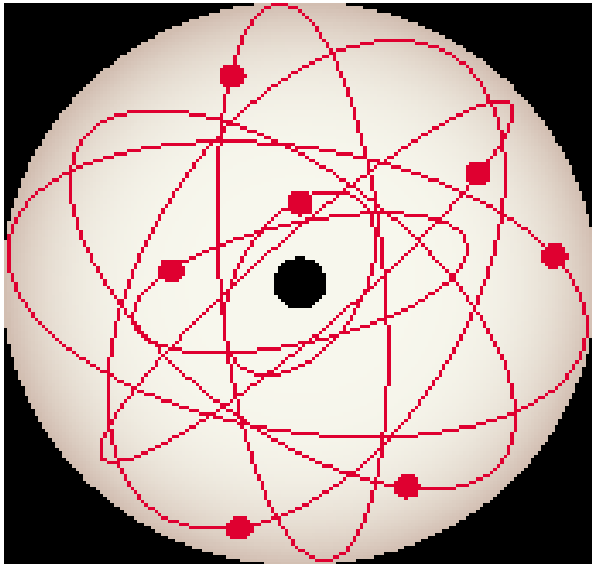
ii) A few particles were deflected sharply and once in a while α -particles would bounce right back to the source.



Why do most of the α -particles pass through straight through the metal foil?

Why did some α -particles appear to rebound from the metal foil?

What factors will affect the extent of scattering of the α -particles?



Rutherford planetary or nuclear atomic model

From the observations, Rutherford concluded that

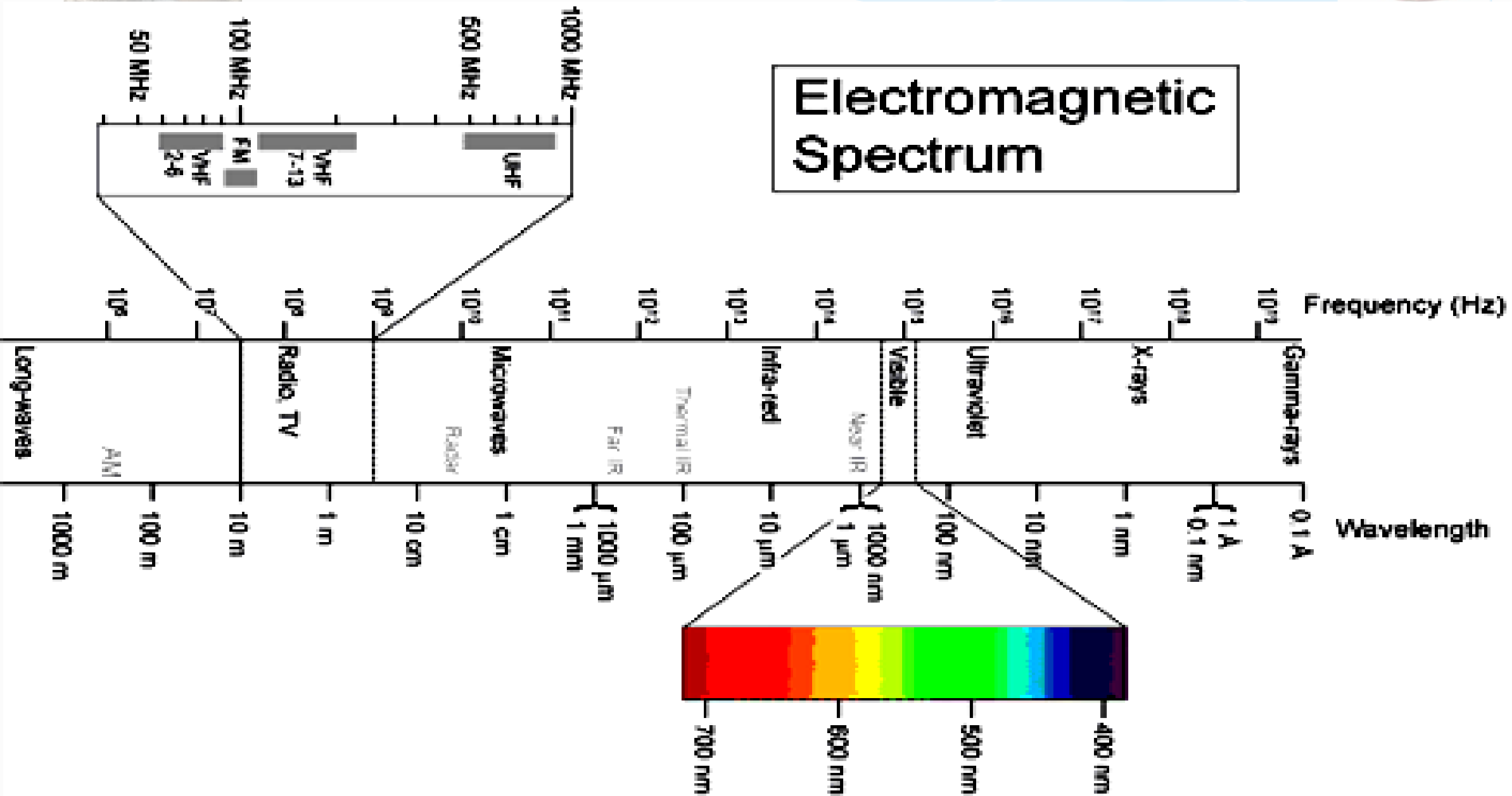
✓The mass of the atom and hence the positive charge was concentrated at the centre of an atom in a tiny core which he called the nucleus.

✓When the positively charged α -particles approach the positively charged nucleus, they get repelled and therefore sharply deflected.

✓Electrons remained in the atom because they are strongly attracted to the positively charged nucleus.



Electromagnetic Spectrum





Much of our current understanding of the electronic structure of atoms has come from analysis of the light emitted or absorbed by substances. To understand the basis for our current model of electronic structure, therefore, we must first learn more about **light**. The light that we can see with our eyes, *visible light*, is type of **electromagnetic radiation**. Electromagnetic radiation carries energy through space in the form of wave.

There are many types of electromagnetic radiation in addition to visible light. These different forms- such as the radio waves that carry music to our radios, the infrared radiation (heat) from the glowing fireplace, and X-rays used in hospital, may *seem* very different from one another, yet they share certain fundamentals characteristics

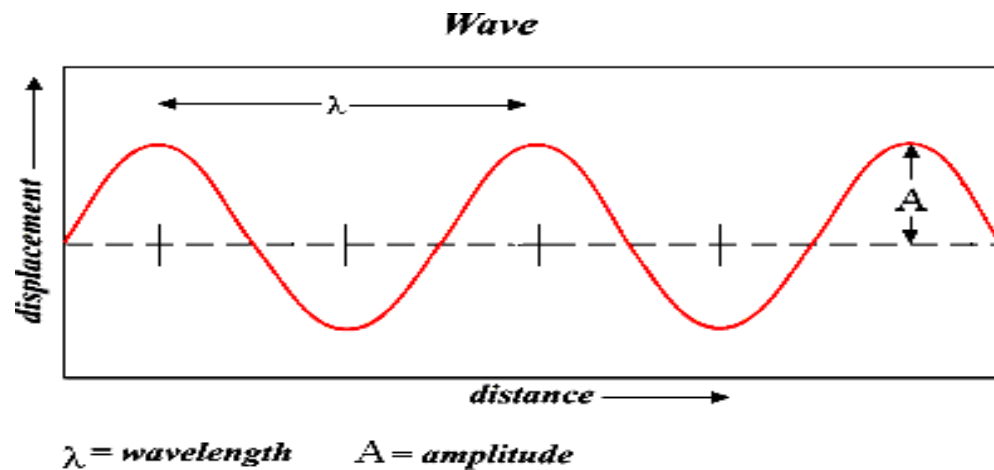
When energy is provided (e.g. as heat or light) to an atom or other species, one or more electrons may be promoted from a ground state level to a higher energy state. This excited state is transient and the electron falls back to the ground state. This produces an *emission spectrum*.



Electromagnetic radiation consists of packets (quanta) of oscillating electric and magnetic fields which are perpendicular to one another and traveling from the source in the form of a harmonic wave.



The harmonic wave is characterized by properties such as frequency, ν wavelength, λ and the wave number, λ^{-1} .



The wavelength λ (**Greek Lambda**), is the distance between two successive crests or troughs in a wave and is expressed in meters, m. The frequency ν (**Greek nu**), is the number of waves passing a point in unit time (usually one second). Its unit is the Hertz; Hz (i.e. cycles per second).



All electromagnetic radiation move through vacuum at a speed of **speed of light, $c = 2.998 \times 10^8 \text{ ms}^{-1}$** .



The wavelength and frequency are related by:

$$c = \lambda \nu$$

Common wavelength units for electromagnetic radiation

Unit	Symbol	Length /m	Type of radiation
Angstrom	Å	10^{-10}	X-ray
Nanometer	nm	10^{-9}	Ultraviolet, visible
Micrometer	μ	10^{-6}	Infrared
Millimeter	mm	10^{-3}	Infrared
Centimeter	cm	10^{-2}	Microwave
Meter	m	10	TV, radio

- What is the frequency of red light whose wavelength is 750 nm?
- What is the wavelength in nm whose frequency is $4.5 \times 10^7 \text{ Hz}$?

The reciprocal of frequency is the period of oscillation, T . It indicates the time taken for one oscillation to pass a point. Hence,

$$T = \frac{1}{\nu}$$

The reciprocal of the wavelength, λ^{-1} , is known as the **wave number**. The wave number indicates the number of waves accommodated in unit length. Thus, since

$$c = \lambda \nu$$

then,

$$\frac{1}{\lambda} = \frac{\nu}{c} = \bar{\nu}$$

The SI unit for wave number is reciprocal metres, m^{-1} , although most of the literature values are given in cm^{-1} . The electric and the magnetic fields interact with matter leading to a spectrum.



Plank's Quantum Theory –Quantized Energy and Photons



Plank's quantum theory state that, atoms and molecules emit or absorb energy only in discrete quantities, like small packets or bundles.

The packets of electromagnetic radiation are known as **quanta** (singular: quantum) or photons.

Each photon has energy, E , given by:

$$E = h\nu$$

where h = Planck's constant = $6.6262 \times 10^{-34} \text{ J s photon}^{-1}$.

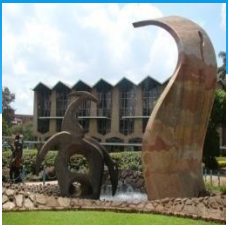
ν = frequency of wave in Hertz

According to Plank's quantum theory, energy is always emitted in whole number multiples of

$h\nu$, for example $2h\nu, 3h\nu, 4h\nu$, but never fractions



- i. For radiation having $\lambda = 2.0 \times 10^{-7} \text{ m}$, calculate:
- (i) $E \text{ (kJmole}^{-1}\text{)}$
 - (ii) ν
 - (iii) λ^{-1}
- ii. Explain why X-ray is more energetic than microwave radiation
- iii. Briefly explain Plank's quantum theory.
- iv. Give two familiar, everyday example to illustrate the concept of quantization



Hydrogen Spectrum



When sodium chloride is heated strongly in a Bunsen, it gives off a **brilliant yellow light**. Some other also emits light when they are heated in this way. Have ever noticed that compounds of one element always emit the same color of light when they behave like this?

SELF-TEST QUESTIONS

1. What colour of light is emitted by:
 - i) Potassium compounds
 - ii) Calcium compounds
 - iii) Copper compounds
2. What are emission spectra?
3. Distinguish line and continuum spectra

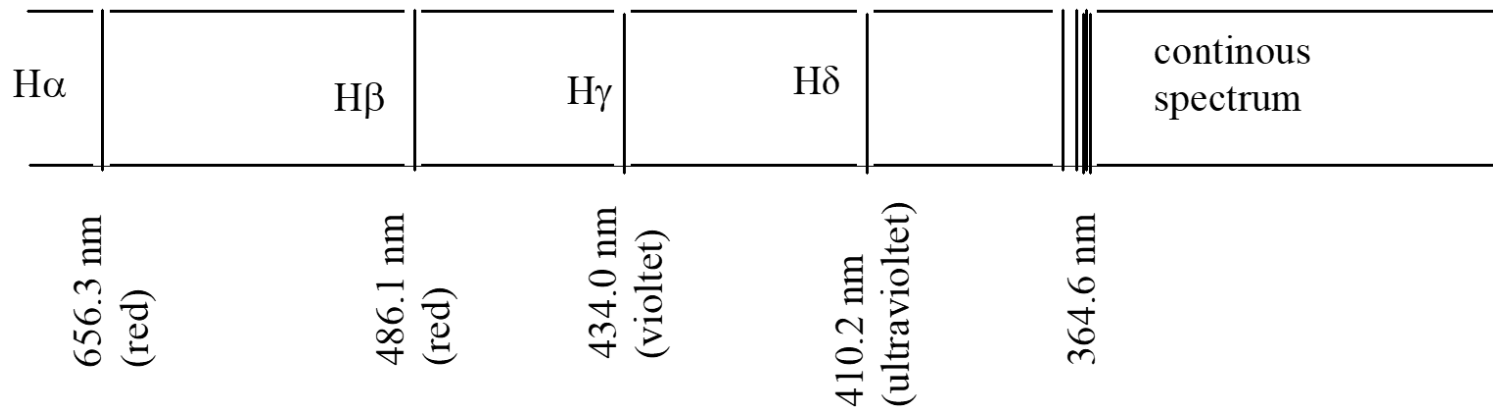


In the visible region, the spectrum of hydrogen was known to consist of 4 main lines

Red 656.3 nm
Blue 486.1 nm
Violet 434.0 and 410.2 nm

Denoted as H_{α} , H_{β} , H_{γ} and H_{δ} .

Near the ultra-violet there were other lines getting closer together converging to a limit of 364.6 nm



The Balmer series of atomic hydrogen excited by electric discharge. The dark lines correspond to the bright emission lines of the spectrum.



According to Balmer (1885, that the wavelength (λ) of the nine lines, then known in this spectrum, could be accurately expressed by a single formula;

$$\lambda = \frac{kn^2}{n^2 - 2^2}$$

Where k is a constant and n is an integer.

The different wavelength corresponding to the lines H_α , H_β , H_γ and H_δ etc. are obtained by putting the integers 3, 4, 5, 6 etc respectively in place of n. This set of line constitutes a spectral series and is known as the **Balmer series**. Expressing in terms of wave numbers, equation (1) is rearranged as

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

Where R is a constant called the Rydberg constant; $R=109677.8 \text{ cm}^{-1}$, n is an integer which can take all values greater than 2, $n=3,4, 5\ldots\infty$.

The hydrogen spectrum was investigated in the far (UV) ultra violet, IR (infra red) regions and revealed that there were few other series such as Lyman (UV region), Paschen and Brackett and Pfund. All these series were found to be in excellent agreement with the general equation

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

This is the universal formula for the atomic hydrogen spectrum. Where n_1 and n_2 are integers somehow defining the energy stages of the atom.

The atomic spectrum of hydrogen atom

Series	Region	n_1	n_2
Lyman	Ultraviolet	1	2, 3, 4, 5..... ∞
Balmer	Visible	2	3, 4, 5, 6..... ∞
Paschen	Near infrared	3	4, 5, 6, 7..... ∞
Brackett	Far infrared	4	5, 6, 7, 8..... ∞
Pfund	Far infrared	5	6, 7, 8, 9..... ∞



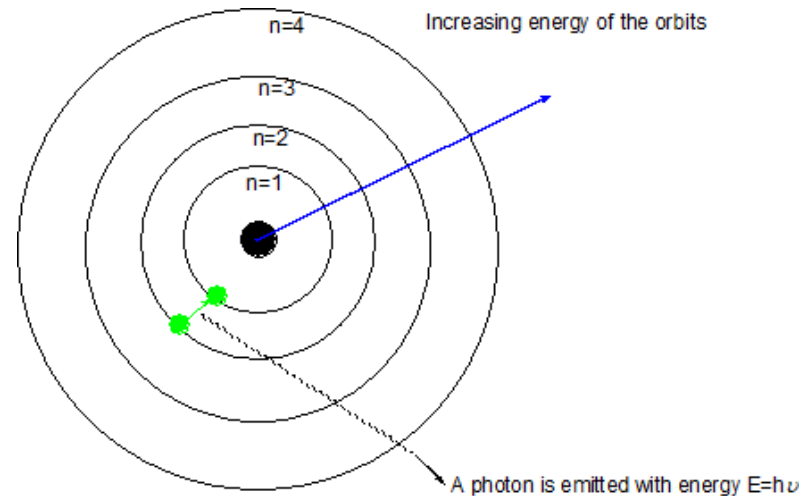
Bohr Model of Atom (1913)

In order to explain the hydrogen spectrum, Bohr made the following **assumptions** governing the behavior of electrons:

1. Electrons revolve in orbits of specific radius around the nucleus without emitting the radiation.
2. Within each orbit, each electron has a fixed amount of energy; electrons in orbits further from the nucleus have greater energies.
3. An electron may 'jump' from one orbit of high energy to another of lower causing the energy difference to be emitted as a photon of electromagnetic radiation such as light.
4. An electron may absorb a photon of radiation and jump from a lower-energy orbit to a higher-energy one.

The energy emitted or absorbed corresponds to the difference in the energy for the **initial and final state of the system**.

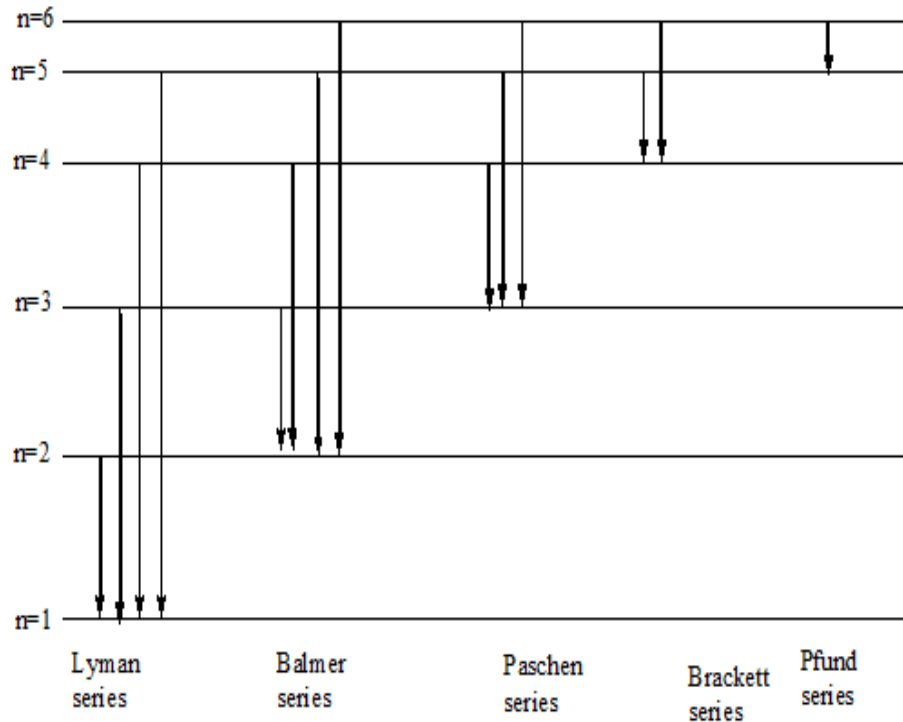
$$E_{\text{final}} - E_{\text{initial}} = \Delta E = h\nu$$



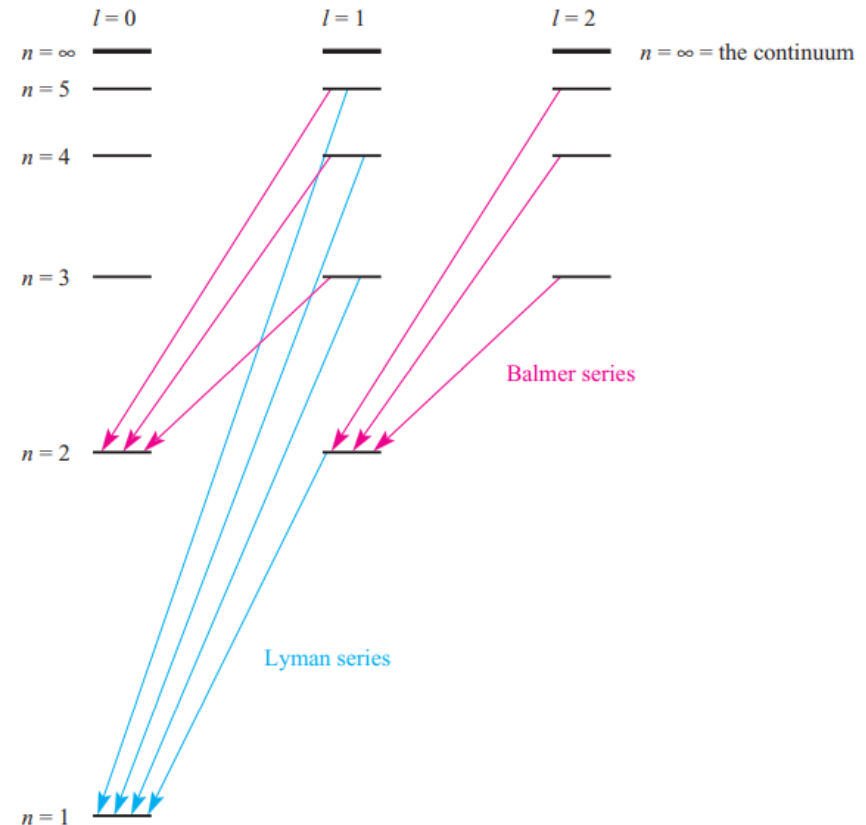
Bohr atomic model



One of the implication of these quantized energy states is that only certain photon energy are allowed when electrons jump down from high to lower levels producing the hydrogen spectrum.



Various series in hydrogen spectrum.



For hydrogen the energy of electron in a n energy level is given by:

$$E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2} = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

or

$$E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

Where

m-mass of electron $9.1079 \times 10^{-31} \text{ kg}$

e-electronic charge $1.602 \times 10^{-19} \text{ C}$

h-Plank's constant $6.626 \times 10^{-34} \text{ J s}$

ϵ_0 -permittivity of free space $8.854188 \times 10^{-12} \text{ C}^2 \text{ s}^2 \text{ kg}^{-1} \text{ m}^{-3}$

Z-nuclear charge

SELF-TEST QUESTIONS

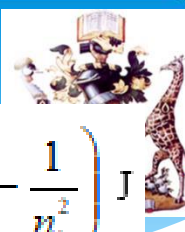
- Calculate the frequency and wavelength of the emitted photon when electron drops from the $n=5$ to the $n=2$ level in a hydrogen atom
- Derive the expression $E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2}$
- Explain why elements produce their own characteristic colours when they emit photons

$$\Delta E = E_{n_2} - E_{n_1}$$

$$\Delta E = -2.18 \times 10^{-18} Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J}$$

or

$$\Delta E = -13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$



Failures of the Bohr Model



While the model was a major step towards understanding the quantum theory of the atom, it is not in fact a correct description of the nature of electron orbit.



Some of the shortcomings of the models are:-

1. It fails to provide any understanding **why some spectral lines are brighter** than others.
2. There is no mechanism for the calculation of **transition probability**.
3. The Bohr model treats electrons as if they were **miniature planet with definite radius and momentum**. This is a direct violation of the **Heisenberg uncertainty principle** which dictates that position and momentum cannot be simultaneously determined. The Bohr model gives us a basic conceptual model of electrons orbit and energies. The precise details of spectra and charge distribution must be left to quantum mechanical calculations as with the **Schrödinger equation**.
5. What are the two longest-wavelength lines (in nanometres) in the Lyman series of the hydrogen spectrum? (Hint: The Lyman series is given by the Balmer–Rydberg equation with $m = 1$ and $n > 1$. The wavelength is greatest when n is smallest. (2 marks)

$$\frac{1}{\lambda} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$



Lecture 2

- ✓ Dual nature of matter and waves
- ✓ The Schrödinger equation and quantum numbers
- ✓ Shapes of atomic orbitals Pauli's exclusion principle
- ✓ Electronic configurations



Atom	Atomic number	Electronic configuration	Atom	Atomic number	Electronic configuration
H	1	1	Na	11	2 8 1
He	2	2	Mg	12	
Li	3	2 1	Al	13	2.8.3
Be	4		Si	14	
B	5		P	15	
C	6	2 4	S	16	
N	7		Cl	17	2 8 7
O	8	2 6	Ar	18	
F	9		K	19	
He	10	2 8	Ca	20	2.8.8.2

In high school you learnt about electronic configuration of atoms.

- The electrons are arranged in energy levels, labeled $n=1, 2, 3, 4$ etc.
- That the maximum number of electrons in each energy level is given by $2(n)^2$.
 - The first energy level can have a maximum of $2(1)^2=2$
 - The second energy level can have a maximum of $2(2)^2=8$
 - The third energy level can have a maximum of $2(3)^2=18$
 - And so on

No explanation was given at that time!

Quantum Model of an Atom



A fundamental concept of quantum mechanics is that all matter has wave and particle-like properties.

De Broglie Equation- Dual Nature of Matter

From Einstein relativity theory, the fundamental law relating energy E , rest mass m_0 and momentum p of a particle is;

$$\left(\frac{E}{c}\right)^2 = p^2 + m_0^2 c^2$$

The rest mass of a photon is zero, so $m_0 = 0$ and therefore momentum is;

$$p = \frac{E}{c}$$

From Plank's quantum theory;

$$E = h \nu$$



From Plank's quantum theory



$$E = h\nu$$

$$E = h\nu = mc^2$$

λ = wavelength of the moving particle
 h : Planck's constant 6.626×10^{-34} J.s; m : mass of the moving particle.

That is, all particles in motion have a wavelength that depends on the particle's momentum. The larger the momentum, the smaller the wavelength.

Therefore; $p = mc$

But
$$p = mc = \frac{h\nu}{c} = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{p}$$

Quiz

What is the de Broglie wavelength (in meters) of a baseball weighing 145 g and traveling at 156 km/h? (2 marks)

Derive the Louis de Broglie equation and state the explanations given to the questions arising from the postulates in the Bohr atom that restricted electrons to orbiting the nucleus at fixed distances. (4 marks)



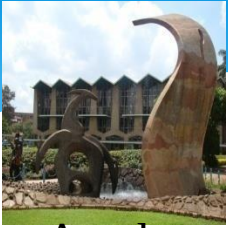
Heisenberg Uncertainty Principle

- ✓ Due to the dual nature of matter, both particles and wavelike properties, it is impossible to simultaneously determine both the position and momentum of an object especially as small as an electron.
- ✓ The position and momentum of a particle cannot be simultaneously measured with high precision.
- ✓ There is a minimum for the product of the uncertainties of these two measurements. There is likewise a minimum for the product of the uncertainties of the energy and time.

$$\Delta x \Delta p \geq \frac{h}{2\pi}$$

Where Δx = uncertainty in position

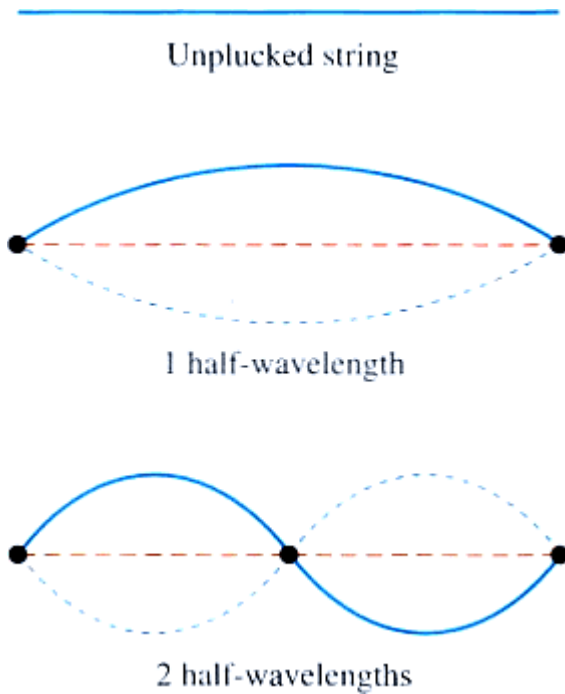
Δp = uncertainty in momentum



Wave-Like Property of an Electron



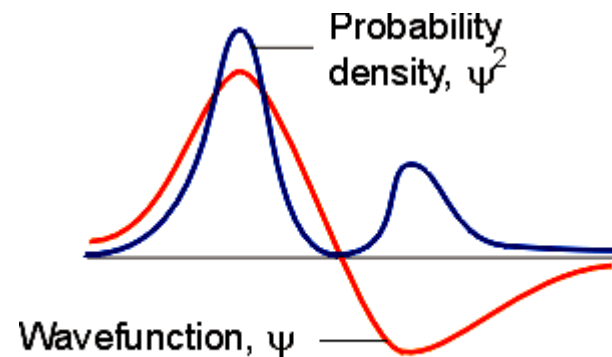
An electron in an atom is viewed as a standing wave. Some common examples of standing waves are: a string attached at both ends to produce a musical tone in a guitar and violin.



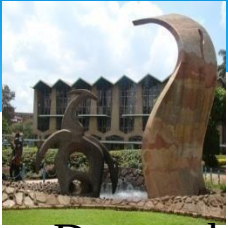
Standing wave

We use the mathematical function Ψ (Greek Psi), called wavefunction to describe the wave-like behavior of an electron in a region of space called the atomic orbital.

- Ψ has positive and negative amplitudes, like any wave.



Figure



- Due to the wave-like behavior of an electron, we cannot pin down its position and momentum at the same time (Heisenberg Uncertainty Principle).
- In other words, we do not know the **exact location** of an electron and how it moves from one spot to another in an atom.
- Instead, we can only **speak of the probability of finding an electron in a given volume of space**. This is given by the function Ψ^2 .

Schrödinger Wave Equation

Developed a partial differential equation to describe the behavior of an electron around an H atom. He received Nobel Prize in 1933.

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

There are many solutions to the Schrödinger's equation. Each solution is represented by the wavefunction, Ψ , which describes an atomic orbital.

Atomic orbitals

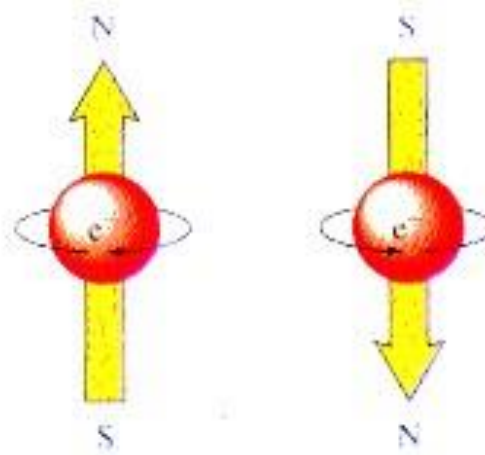


An atomic orbital is defined as the boundary surface encloses 95% of the electron density for a wavefunction Ψ

Each atomic orbital (Ψ) may be uniquely defined by a set of three quantum numbers:

- n (the principal quantum number);
- l (the orbital angular quantum number) and
- m_l (the magnetic quantum number).

Since an electron behaves as if it is spinning about an axis, a spin quantum number m_s is also required to uniquely describe an electron in an atomic orbital.



Spinning electron

Quantum Numbers and their Properties

Symbol	Name	Allowed Values	Properties
N	Principal Quantum Number	$n = 1, 2, 3, 4, \dots$	- determines the size and energy of an atomic orbital - As n increases, the number of allowed orbitals increases, the size and energies of those orbitals also increase.
L	Orbital Quantum Number	$l = 0, 1, 2, \dots, n-1$	Describe the shape of an atomic orbital
m_l	Magnetic Quantum Number	$m_l = -l, \dots, -(l-1), 0, 1, 2, \dots +l$	Describe the directionality of an atomic orbital
m_s	Spin Quantum Number	$m_s = -\frac{1}{2}, +\frac{1}{2}$	Describes the orientation of the electron spin in space

Orbitals with different l values are known by the following labels:

L	0	1	2 3 4 5 ...
Label	<i>s</i>	<i>p</i>	<i>d f g h ...</i>

Allowed values enable us to know the maximum number of electrons in each energy

level



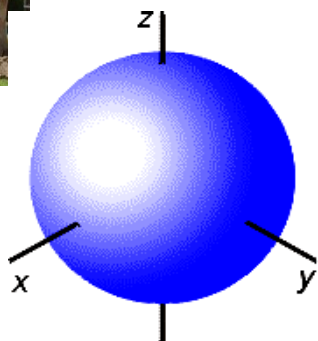
n	<i>l</i>	m_l and m_s					
1	0	0					
		$\uparrow\downarrow$					2 electrons in 1s orbital
							Total of 2 electrons in the FIRST energy level
2	0	0					
		$\uparrow\downarrow$					2 electrons in 2s orbital
	1	-1	0	1			
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$			
							Total of 8 electrons in the SECOND energy level; 2 in s orbital and 6 in p orbitals
3	0	0					
		$\uparrow\downarrow$					2 electrons in 3s orbital
	1	-1	0	1			
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$			
	2	-2	-1	0	1	2	
		$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	10 electrons in 3d orbital
							Total of 18 electrons in the THIRD energy level; 2 in s orbital, 6 in p orbitals and 10 in d orbitals

Using the same argument, show the maximum number of electrons for $n=4$.

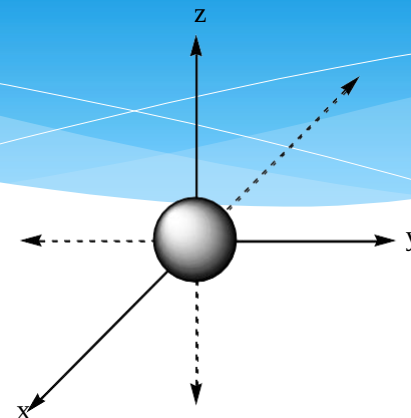
Pictorial Representation of Atomic Orbitals



S-orbital



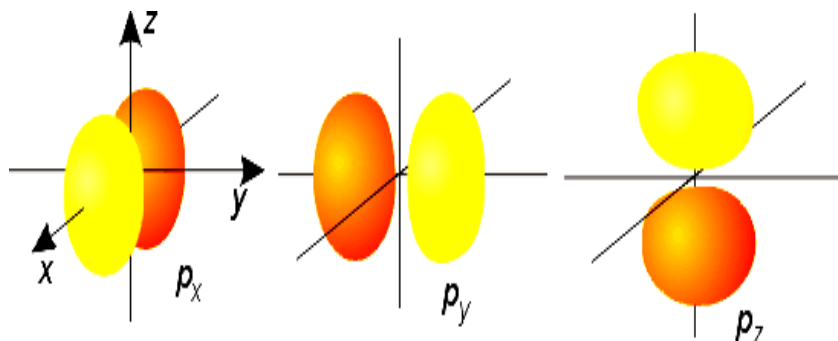
Boundary surface diagram



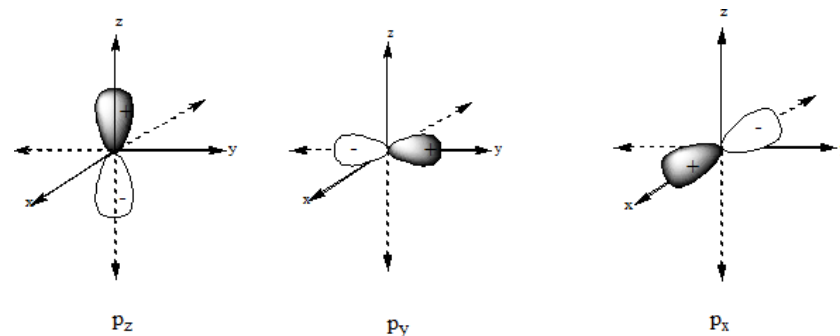
s orbital

Two dimensions

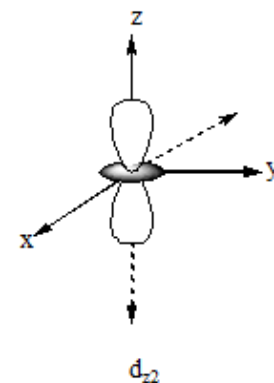
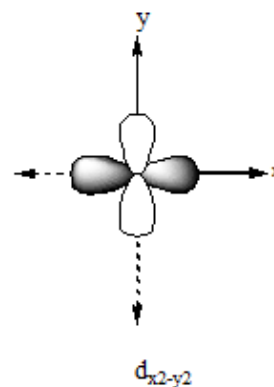
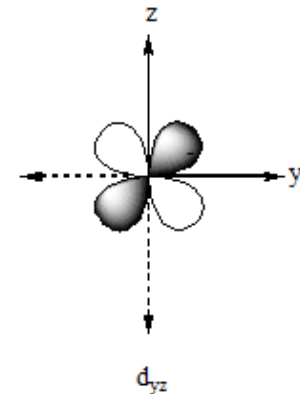
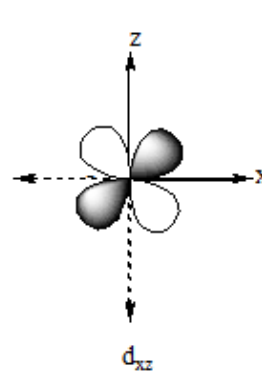
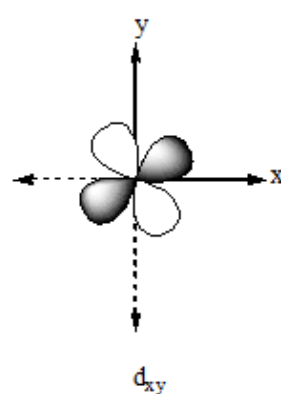
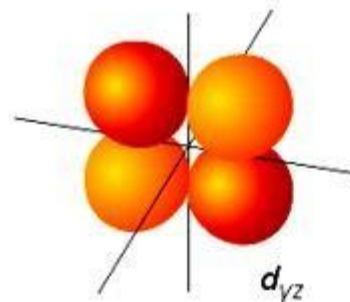
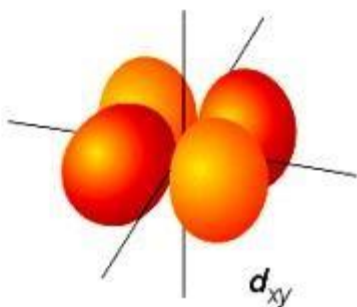
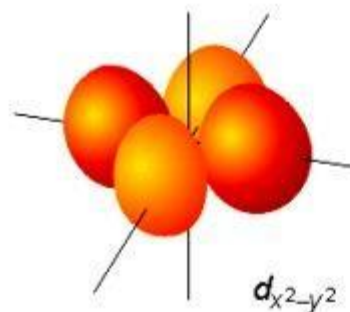
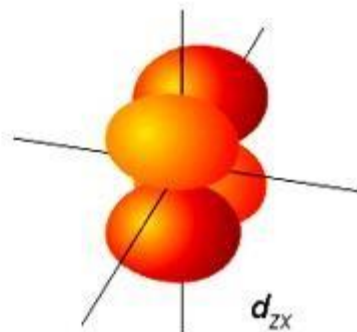
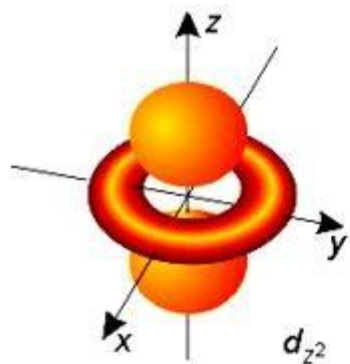
P-orbital



Boundary surface diagram



Two dimensions



The shaded lobe has a positive sign
Note the labelling of the axes

2-dimension

Boundary surface diagram of d orbitals. The lightly shaded lobe has a positive amplitude; the more darkly shaded lobe is negative

The Pauli Exclusion Principle



This principle states that *no two electrons in an atom can have the same four quantum numbers*. If two electrons in an atom should have the same n , l and m_l values (that is, these two electrons are in the same orbital), then they must have different values of m_s . In other words, **only two electrons may exist in the same atomic orbital, and these electrons must have opposite spins**.

Consider the helium atom, which has two electrons. The three possible ways of placing two electrons in the $1s$ orbital are as follows:

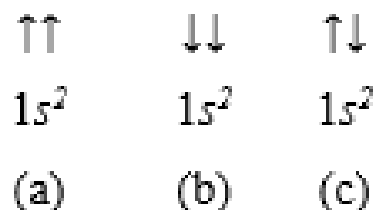


Diagram (a) and (b) are ruled out by the Pauli Exclusion Principle.



Electronic Configuration



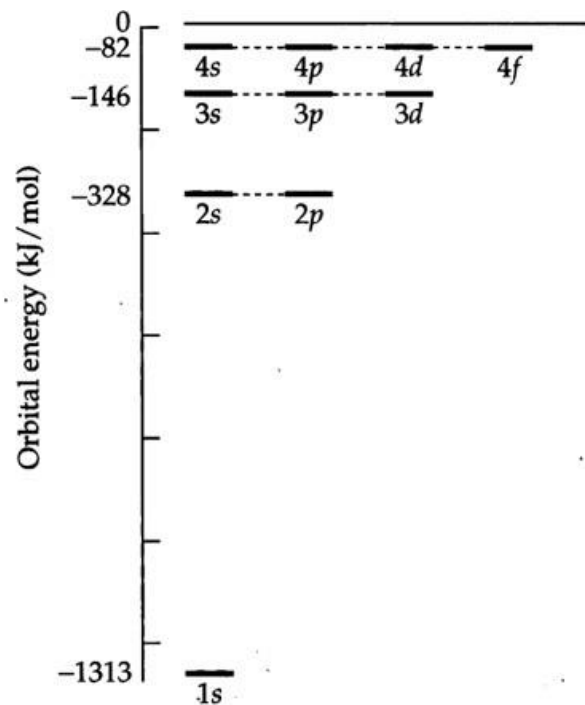
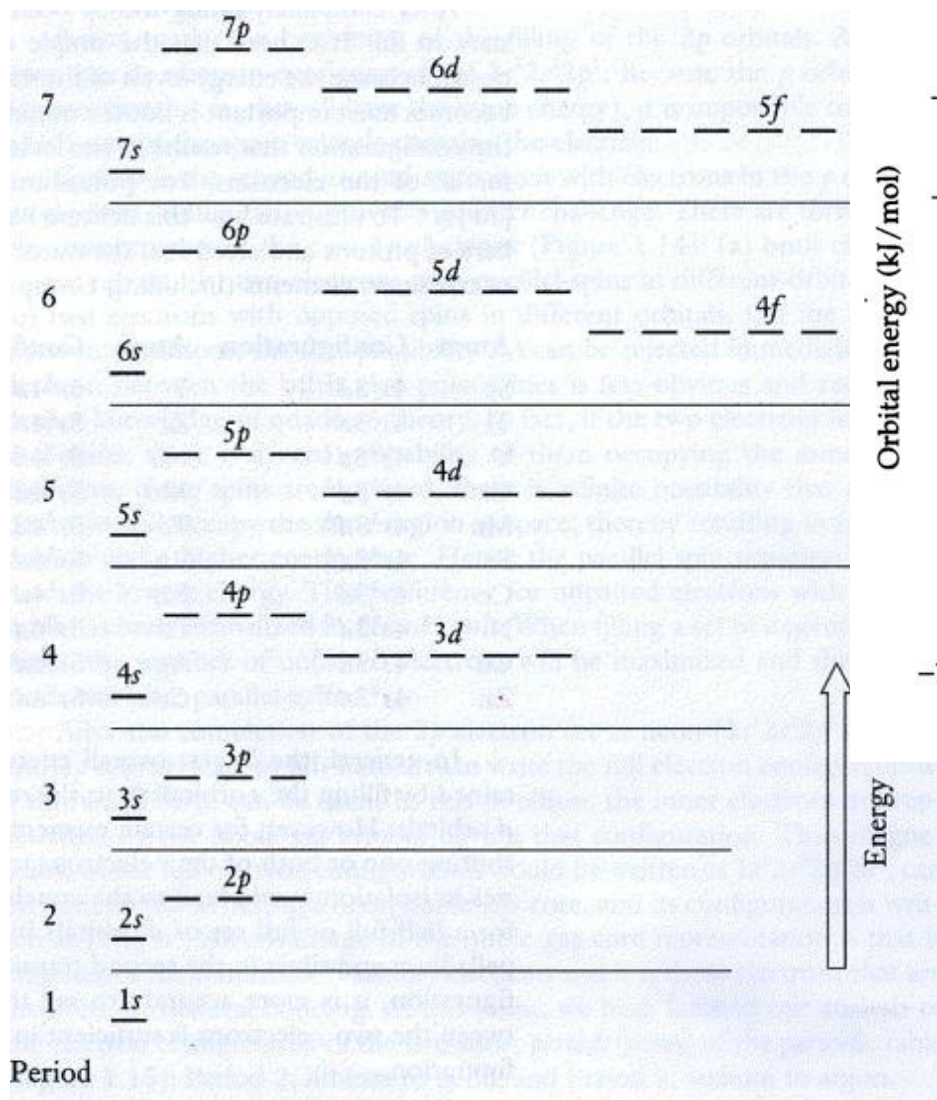
Thus the helium atom has the following configuration:



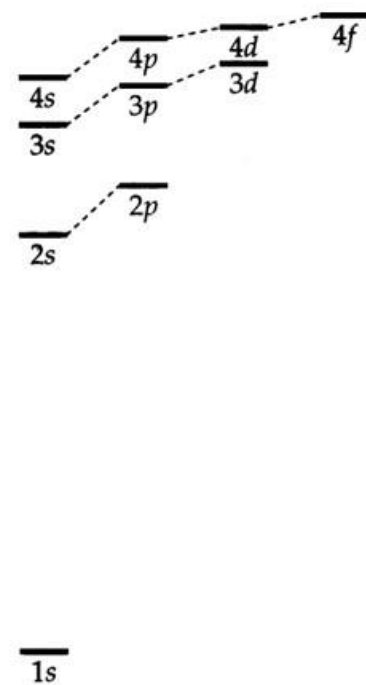
Note that $1s^2$ is read “one s two”, not “one s squared”

Energy Levels in Hydrogen and Other Atoms

1. For H, the energy of the electron only depends on n , principal quantum number.
The s , p , d , f orbitals have the same energy.
2. In the ground state, the electron in the H atom resides on the $1s$ orbital.
3. For other atoms with more than one electron, electron-electron repulsion and shielding play a role so **energy depends on angular momentum quantum number l as well as the principal quantum number n .**



(a) Hydrogen



(b) Multielectron atoms

The energies of the atomic orbitals in multielectron atoms are arranged in the order:

$$ns < np < nd < nf$$



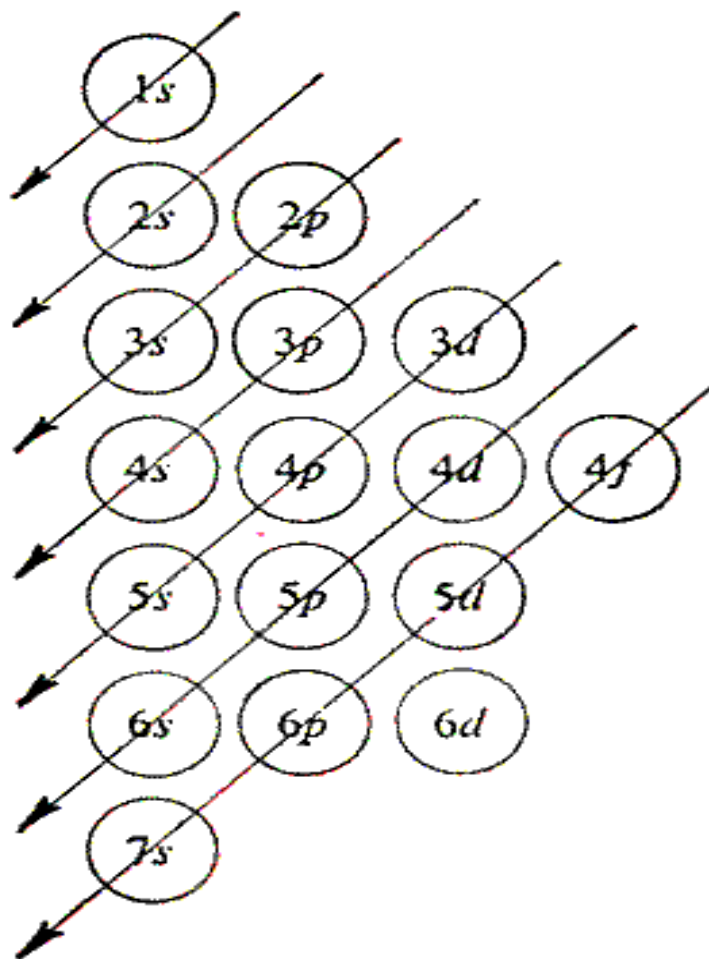
Aufbau Principle



The German word *Aufbau* means ‘building up’)

- A set of rules guiding the filling order of orbitals.
- 1 Fill in electrons in the lowest energy orbital first.
 - 2 Only two electrons with opposite spins per orbital (because Pauli exclusion principle says that no two electrons in an atom can have the same four quantum numbers).
 - 3 If two or more degenerate orbitals (i.e. orbitals with the same energy) are available, electrons go into each degenerate orbital with parallel spins until the orbitals are half-fill (Hund's rule).

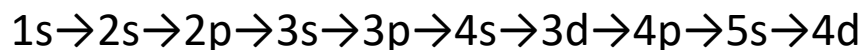
Order of Orbital Filling in Poly-electronic Atoms



The order in which atomic orbitals are filled in a many electron atoms.

Start with the 1s orbital and move downward, following the direction of the arrows.

Thus the order goes as follows



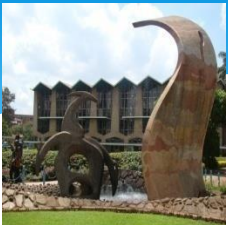
Filling of electrons for the first 30 elements



Atom	Electronic configuration	Atom	Electronic configuration
H	$1s^1$	S	$1s^2 2s^2 2p^6 3s^2 3p^4$
He	$1s^2$	Cl	$1s^2 2s^2 2p^6 3s^2 3p^5$
Li	$1s^2 2s^1$	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$
Be	$1s^2 2s^2$	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
B	$1s^2 2s^2 2p^1$	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
C	$1s^2 2s^2 2p^2$	Sc	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
N	$1s^2 2s^2 2p^3$	Ti	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
O	$1s^2 2s^2 2p^4$	V	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
F	$1s^2 2s^2 2p^5$	Cr	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Ne	$1s^2 2s^2 2p^6$	Mn	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
Na	$1s^2 2s^2 2p^6 3s^1$	Fe	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
Mg	$1s^2 2s^2 2p^6 3s^2$	Co	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
Al	$1s^2 2s^2 2p^6 3s^2 3p^1$	Ni	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
Si	$1s^2 2s^2 2p^6 3s^2 3p^2$	Cu	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$
P	$1s^2 2s^2 2p^6 3s^2 3p^3$	Zn	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

4	Be	[He]2s ²	41	Nb	[Kr]5s ² 4d ³	77	Ir	[Xe]6s ² 4f ¹⁴ 5d ⁷
5	B	[He]2s ² 2p ¹	42	Mo	[Kr]5s ¹ 4d ⁵	78	Pt	[Xe]6s ¹ 4f ¹⁴ 5d ⁹
6	C	[He]2s ² 2p ²	43	Tc	[Kr]5s ² 4d ⁵	79	Au	[Xe]6s ¹ 4f ¹⁴ 5d ¹⁰
7	N	[He]2s ² 2p ³	44	Ru	[Kr]5s ² 4d ⁶	80	Hg	[Xe]6s ² 4f ¹⁴ 5d ¹⁰
8	O	[He]2s ² 2p ⁴	45	Pd	[Kr]5s ² 4d ⁷	81	Tl	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ¹
9	F	[He]2s ² 2p ⁵	46	Ag	[Kr]5s ² 4d ⁸	82	Pb	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ²
10	Ne	1s ² 2s ² 2p ⁶	47	Ag	[Kr]5s ¹ 4d ¹⁰	83	Bi	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ³
11	Na	[Ne]3s ¹	48	Cd	[Kr]5s ² 4d ¹⁰	84	Po	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁵
12	Mg	[Ne]3s ²	49	In	[Kr]5s ² 4d ¹⁰ 5p ¹	85	At	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁵
13	Al	[Ne]3s ² 3p ¹	50	Sn	[Kr]5s ² 4d ¹⁰ 5p ²	86	Rn	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁶
14	Si	[Ne]3s ² 3p ²	51	Sb	[Kr]5s ² 4d ¹⁰ 5p ³	87	Fr	[Rn]7s ¹
15	P	[Ne]3s ² 3p ³	52	Te	[Kr]5s ² 4d ¹⁰ 5p ⁴	88	Ra	[Rn]7s ²
16	S	[Ne]3s ² 3p ⁴	53	I	[Kr]5s ² 4d ¹⁰ 5p ⁵	89	Ac	[Rn]7s ² 6d ¹
17	Cl	[Ne]3s ² 3p ⁵	54	Xe	[Kr]5s ² 4d ¹⁰ 5p ⁶	90	Th	[Rn]7s ² 6d ²
18	Ar	[Ne]3s ² 3p ⁶	55	Cs	[Xe]6s ¹	91	Pa	[Rn]7s ² 5f ² 6d ¹
19	K	[Ar]4s ¹	56	Ba	[Xe]6s ²	92	U	[Rn]7s ² 5f ³ 6d ¹
20	Ca	[Ar]4s ²	57	La	[Xe]6s ² 5d ¹	93	Np	[Rn]7s ² 5f ⁴ 6d ¹
21	Sc	[Ar]4s ² 3d ¹	58	Ce	[Xe]6s ² 4f ¹ 5d ¹	94	Pu	[Rn]7s ² 5f ⁶

The Ground State Electron Configuration of the Elements*



*The symbol [He] is called the *helium core* and represents $1s^2$. [Ne] is called the *neon core* and represents $1s^2 2s^2 2p^6$. [Ar] is called the *argon core* and represent $[Ne] 3s^2 3p^6$. [Kr] is called the *krypton core* and represents $[Ar] 4s^2 3d^{10} 4p^6$. [Xe] is called the *xenon core* and represents $[Kr] 5s^2 4d^{10} 5p^6$. [Rn] is called the *radon core* and represents $[Xe] 6s^2 4f^{14} 5d^{10} 6p^6$.

UNIVERSITY EXAMINATIONS 2020/2021
FIRST YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (MECHANICAL, CIVIL, PETROLEUM AND BIOSYSTEMS
ENGINEERING)

FEB 105/FME 151/FPE 151/FCE 181: CHEMISTRY IA

TIME: 2 HOURS

USEFUL CONSTANTS:

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$$

$$N_0 = 6.023 \times 10^{23} \text{ mol}^{-1}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$h = \text{Planck's constant} = 6.626 \times 10^{-34} \text{ J.s}$$

$$m = \text{mass of electron } 9.1079 \times 10^{-31} \text{ kg}$$

$$e = \text{electronic charge } 1.602 \times 10^{-19} \text{ C}$$

$$R_H = 109677.8 \text{ cm}^{-1} / 1.097 \times 10^{-2} \text{ nm}^{-1}. R = \text{Gas}$$

$$\text{constant} = 8.314 \text{ J/K.mol}$$

INSTRUCTIONS:

ANSWER QUESTION ONE (COMPULSORY) AND THREE OTHER QUESTIONS

QUESTION ONE (25 MARKS)

1. What are the four quantum numbers, and what does each specify? (2 marks)
2. Draw orbital-filling diagrams for atoms with the following atomic numbers. Show each electron as an up or down arrow and use the abbreviation of the preceding noble gas to represent inner-shell electrons.
 - A. $Z = 28$
 - B. $Z = 47$ (2 marks)